British Airways Passenger CSAT Dashboard

Portfolio Project Overview

British Airways (BA), one of the world's leading flag carriers, processes vast amounts of customer feedback across global routes, aircraft types, and service touchpoints. The airline's Public Relations (PR) and Customer Experience teams require actionable, data-driven insights to share with flight operations, maintenance, and executive leadership.

As part of this end-to-end data analytics portfolio project, an interactive **British Airways Passenger CSAT Analytics Suite** was built to translate raw, multi-channel passenger survey and review data into strategic operational intelligence.

Dashboard Showcase & Analysis Breakdown

1. Executive Dashboard Overview

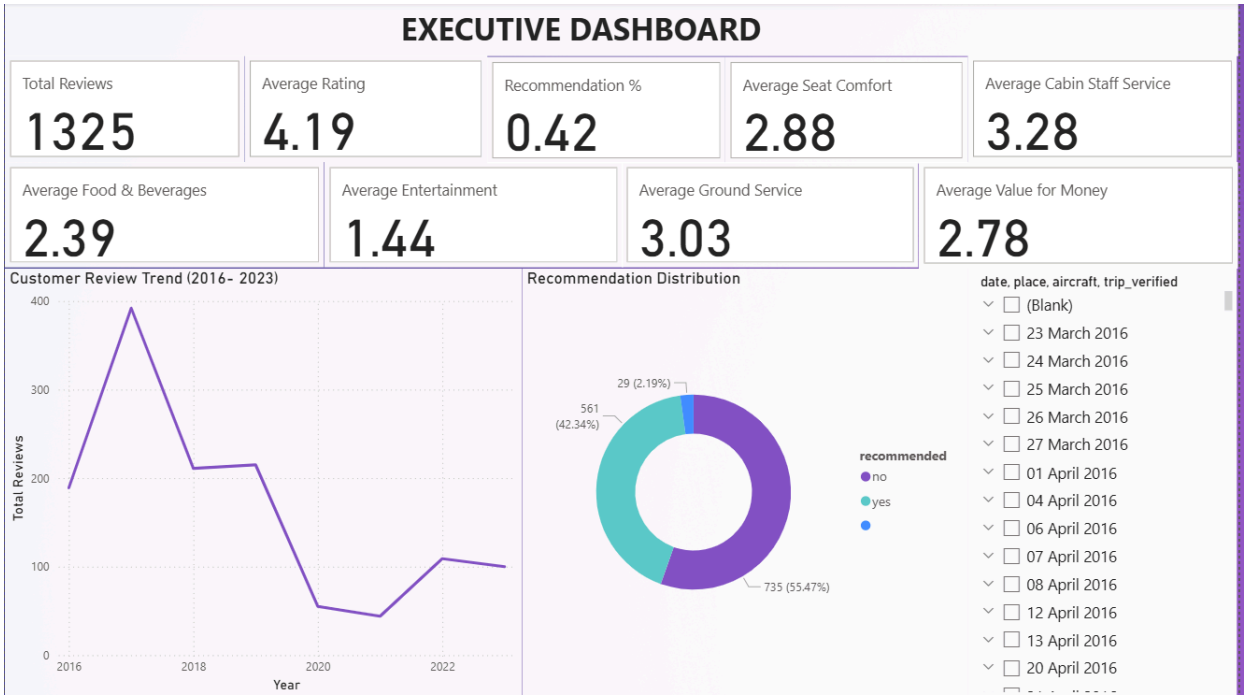


Figure 1: Macro Executive Dashboard displaying high-level KPI scorecards, 8-year review volume trends, and overall recommendation distribution.

2. Customer & Aircraft Review Analysis

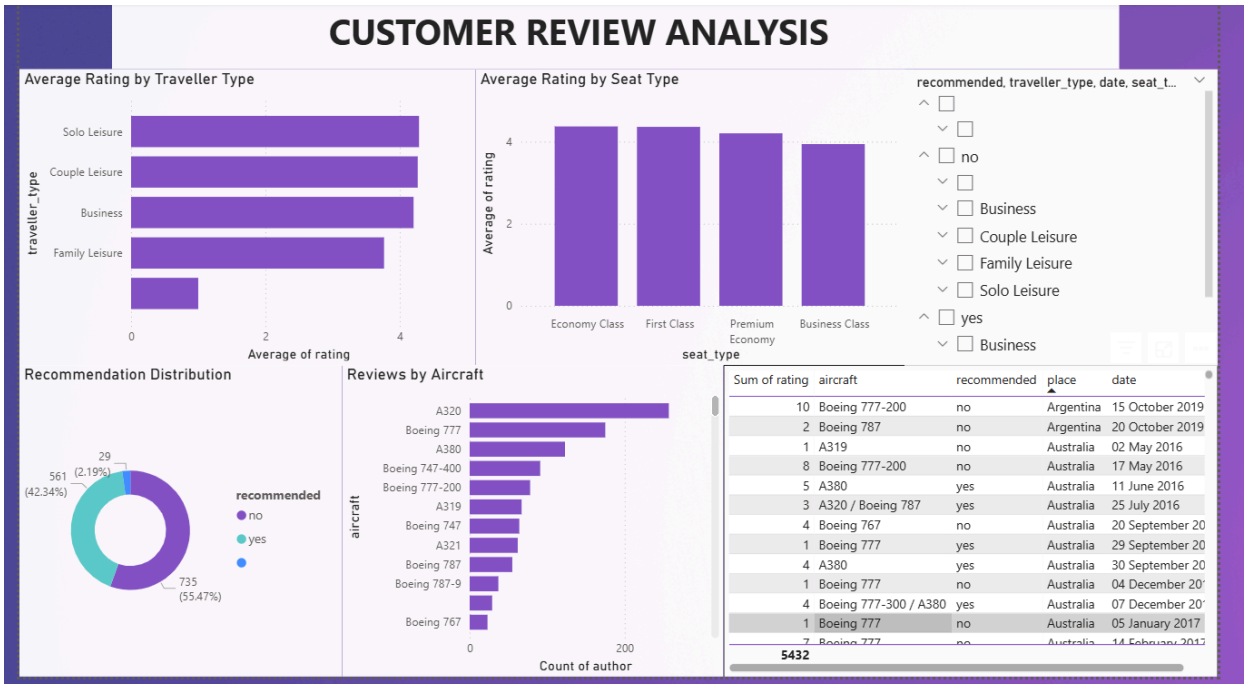


Figure 2: Granular breakdown of average ratings across traveler types, seat categories, aircraft fleets (A320, Boeing 777, A380), and granular transaction logs.

3. Geographic & Time Analysis

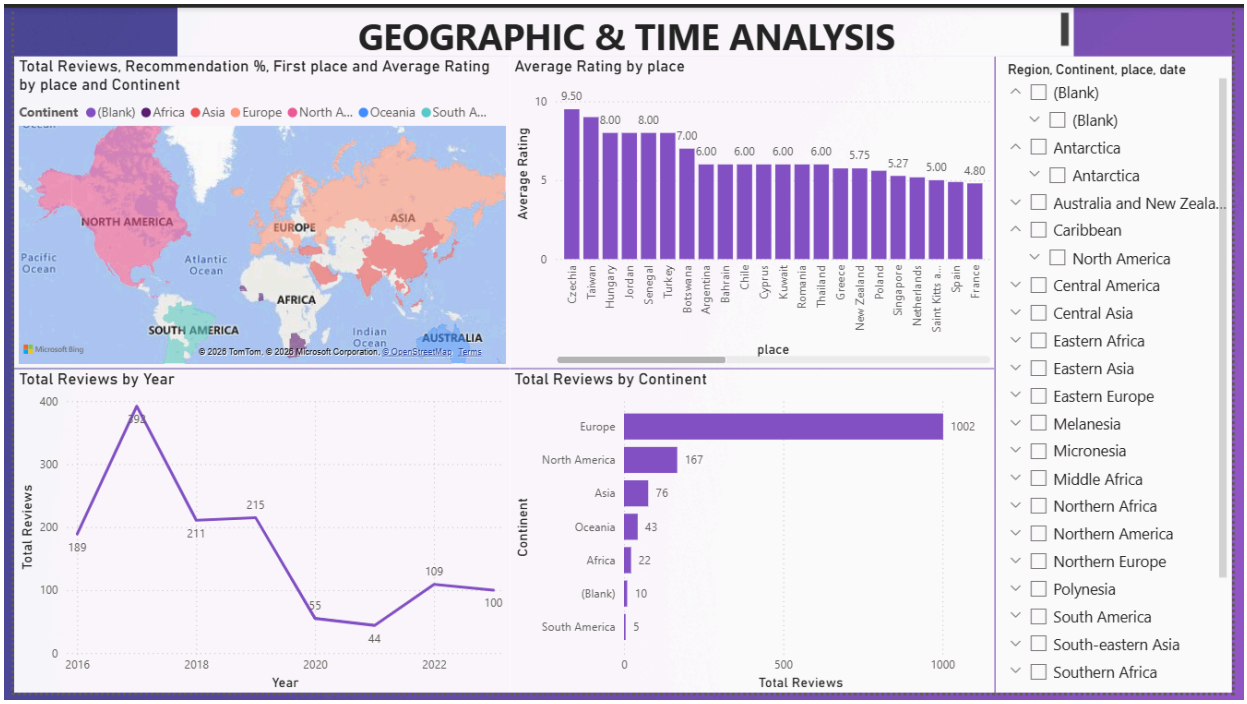


Figure 3: Global spatial mapping, country-level satisfaction rankings, review volume by continent, and timeline distribution from 2016 to 2023.

4. Service Quality Breakdown

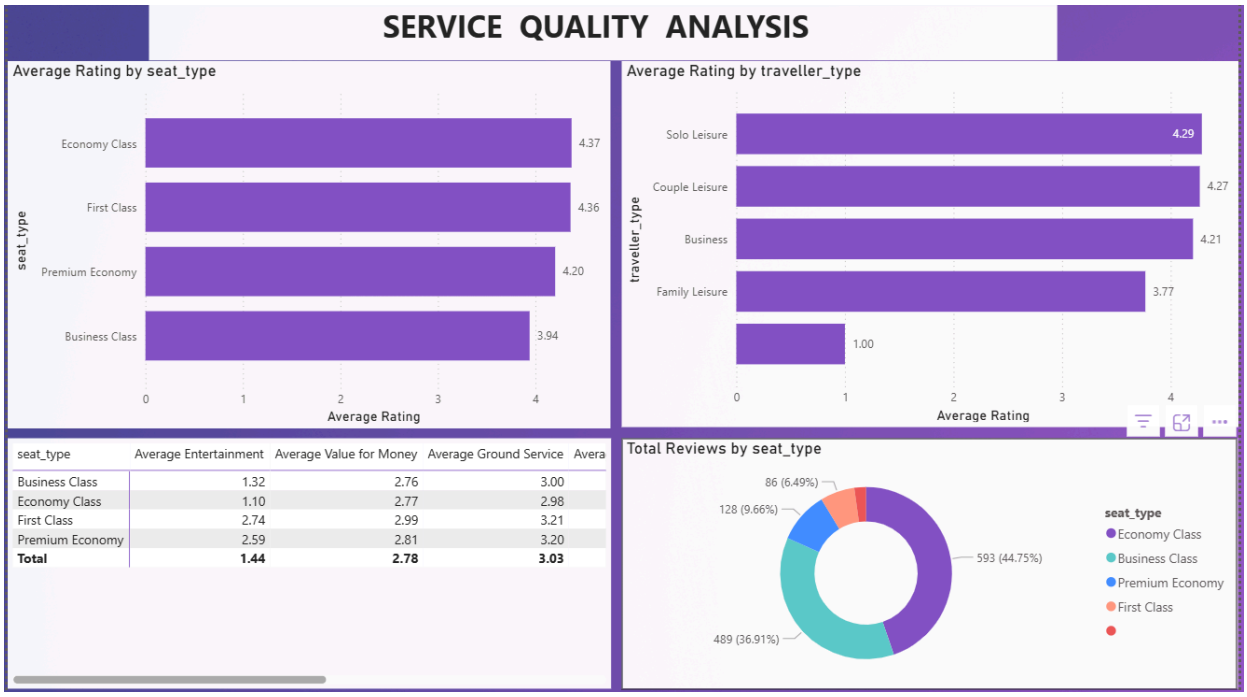


Figure 4: Touchpoint deep dive isolating Seat Comfort, Cabin Staff Service, Food & Beverage, Ground Service, and In-Flight Entertainment ratings by seat class.

Business Problem Scenarios & Objectives

The primary objective is to analyze CSAT survey data to uncover root causes of passenger dissatisfaction and provide continuous improvement strategies across airline operations:

- **Identify Sentiment Patterns:** Unveil core drivers of overall satisfaction and recurring friction points across specific touchpoints (e.g., Seat Comfort, Food & Beverage, Cabin Staff, Ground Service, In-Flight Entertainment).
- **Operational Optimization:** Deliver insights tailored to operational variables including aircraft model (A320, Boeing 777, A380), travel class (Economy, Premium Economy, Business, First), passenger demographic/traveler type (Solo Leisure, Business, Couple, Family), and flight routes.
- **Dynamic Metric Analysis:** Enable stakeholders to seamlessly switch analytical perspectives across multiple rating dimensions without visual clutter.

Target Stakeholders & Decision Matrix

Stakeholder Role	Primary Analytical Focus	Key Strategic Decisions & Actions
VP & Managers – Flight Operations	Cabin Staff Service, Ground Service performance, Route-specific CSAT, Passenger Class trends	Resource allocation, crew training modules, ground handling SLA enforcement, and boarding process optimization.
VP & Managers – Maintenance & Engineering	Seat Comfort ratings, In-Flight Entertainment (IFE) scores, Fleet/Aircraft-specific feedback	Prioritizing cabin retrofits, scheduling deferred hardware maintenance, and guiding fleet lifecycle planning.
Business & Product Analysts	Overall CSAT Rating, Value for Money, dynamic metric correlations	Pricing strategy adjustments, cabin reconfiguration ROI analysis, and competitive benchmarking.

Skills, Tools & Data Architecture

Core Technologies & Applications Used

- **Data Visualization & Analytics:** Power BI / Tableau Desktop & Tableau Public
- **Data Processing & Analysis:** Python (Pandas) / Microsoft Excel
- **Design & Methodology:** Subject Matter Expertise (SME) in CX/CSAT metrics, Business Requirement Gathering, UI/UX Dashboard Layout Design, and User Acceptance Testing (UAT)

Data Pipeline & Data Modeling Strategy

1. **Data Sources:**
    - `ba_reviews.csv`: Raw passenger reviews, flight metadata, categorical tags, and individual 1–10/1–5 ratings.

- `Countries.csv`: Spatial metadata mapping origin countries to ISO alpha codes, regions, and geographic coordinates.

2. **Data Cleaning & Transformation:**
    - **Data Type Conversion & Formatting:** Standardized date fields (`Review_Date`, `Date_Flown`) and cast rating metrics into numerical floats.
    - **Missing & Null Value Management:** Maintained strict null handling for ratings to prevent artificially downward-skewed averages.
    - **Text Parsing & Tagging:** Extracted verification prefixes (e.g., *"Trip Verified |"*) into explicit boolean flags (`Verification Status`).
    - **Outlier & Frequency Analysis:** Cleaned duplicate feedback records using composite key logic (`Passenger Name` + `Review Date` + `Flown Date`).
3. **Data Modeling (Star Schema / E-R Diagram):**
    - Configured Logical Layer relationships between `ba_reviews` (Fact Table) and `Countries` (Dimension Table) using standardized country name keys (`place`).

Strategic Recommendations & Actionable Insights

Key Operational Takeaways

1. **In-Flight Entertainment & Technology Overhaul:**
    - **Insight:** In-Flight Entertainment receives the lowest average rating across all cabins (**1.44 / 5.00**), with Business Class recording a particularly low **1.32**.
    - **Action:** Direct Engineering and Fleet Maintenance teams to prioritize immediate hardware repairs, screen responsiveness upgrades, and media library updates on long-haul aircraft.
2. **Business Class Value Proposition Alignment:**
    - **Insight:** Business Class holds the lowest overall average rating (**3.94 / 10.00**) among cabin types, impacted by below-average ratings in Ground Service (**3.00**) and Value for Money (**2.76**).
    - **Action:** Review premium catering quality and streamline priority check-in/baggage handling SLAs at major hub airports to restore perceived value for high-yield passengers.
3. **Family Leisure Travel Enhancement:**

- **Insight:** Family Leisure travelers exhibit lower satisfaction (**3.77 / 10.00**) compared to Solo Leisure (**4.29**) and Couple Leisure (**4.27**).
- **Action:** Optimize family boarding procedures and enhance dedicated cabin staff assistance during long-haul family travel routes.